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Multi-Objective Reinforcement Learning Using Routed Ensembles and Trajectory Attribution

Objective-Conflict Coefficients Quantify Preference Trade-offs, While Routed Ensembles and Kernel-Surrogate Attribution Close the Multi-Objective Efficiency Gap

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Multi-objective reinforcement learning demands a single agent to simultaneously satisfy multiple reward objectives weighted by a user-supplied preference vector—a task made hard when objectives conflict and gradient directions oppose each other. This paper introduces objective-conflict coefficients that formally quantify this conflict, derives matching online regret bounds, and proposes two practical algorithms: a routed ensemble that routes each trajectory to the basis policy with highest expected return under its preference weight, and a trajectory attribution module based on kernel ridge regression that identifies which trajectories are most valuable for which preference objectives.

AT A GLANCE

Problem: A single preference-conditioned policy fails when objectives strongly conflict; the degree of conflict was not previously formalised.

Why it matters: MORL is central to robotics, resource allocation, and fairness-aware learning, where trade-offs among objectives must be navigated at deployment.

Method: Objective-conflict coefficients + routed ensemble policy learning (trajectory routing to K basis policies) + kernel ridge regression trajectory attribution.

Result: Consistent improvements over single-policy and standard ensemble baselines on multi-objective robotic control benchmarks.

Evidence: Multi-objective MuJoCo locomotion benchmarks; formal regret analysis as a function of maximum conflict coefficient.

THE BIG PICTURE

In multi-objective RL (MORL), the agent receives a vector reward r = (r_1,...,r_K) and a preference weight w in the simplex. The goal is to maximize w^T J(pi) across many possible w, learning the entire Pareto front of preference-weighted returns.

Conflict arises when grad J_i and grad J_j point in opposing directions, forcing any gradient step to damage at least one objective. Prior approaches either concatenate w to the state (ignoring conflict) or use hand-crafted ensemble partitions (ignoring data efficiency). This work addresses both gaps.

WHY EXISTING APPROACHES FALL SHORT

Conditioned single policy: Appending w to the state does not prevent gradient conflicts. For strongly conflicting objectives, the single policy converges to a suboptimal compromise rather than the Pareto front.

Naive ensembles: Manually partitioning the preference space into K fixed regions ignores that trajectories generated under one preference weight may be highly informative for another.

No formal conflict measure: Without a quantitative measure, there is no principled way to predict how many basis policies are needed.

CORE MECHANISM

Routed Ensemble Policy Learning maintains K basis policies {pi_1,...,pi_K} and iterates: (1) Collect trajectories by sampling basis policies under current preference weights. (2) Route: assign each trajectory tau_j to basis policy k* = argmax_k w_j^T J_k(pi_k). (3) Update: apply PPO to each pi_k using only routed trajectories.

Trajectory Attribution uses a kernel ridge regression surrogate f_hat(pi, w; D) to estimate policy performance as a function of the training trajectory set D, enabling curriculum sampling of high-value trajectories.

TECHNICAL FORMULATION

Objective-Conflict Coefficient for objectives i and j:

c_ij(pi) = -grad_theta J_i . grad_theta J_j / (||grad J_i|| ||grad J_j||)

Regret bound: Regret(T) = O(sqrt(T * c_max)) where c_max is the max conflict.

Kernel surrogate: f_hat(w; D) = y^T (K + lambda I)^{-1} k(w)

where K_ij = kappa(phi(tau_i), phi(tau_j)) and y_j = w_j^T J(pi).

LEARNING OR INFERENCE PROCEDURE

Each training step: (1) Sample preference weights from the prior. (2) Collect trajectories by rolling out each basis policy. (3) Compute attribution scores; upsample high-attribution trajectories. (4) Route trajectories to basis policies and apply PPO updates.

At test time, given preference w*, evaluate J_k(pi_k) for each basis policy and return the best.

WHAT THE GUARANTEE SAYS

The regret bound O(sqrt(T * c_max)) provides an instance-dependent characterisation: MORL is hard when objectives conflict strongly and easy when they are compatible. The routed ensemble achieves this rate for any K >= c_max / epsilon. This is the first formal characterisation of MORL hardness in terms of objective geometry.

EXPERIMENTAL FINDINGS

Multi-objective MuJoCo: Routed ensemble outperforms conditioned single-policy baselines by 8–15% in hypervolume metric across 4 multi-objective task suites.

Vs. naive ensemble: Attribution-based curriculum sampling improves sample efficiency by ~30% (same hypervolume with fewer environment interactions).

Conflict coefficient: Empirical c_max correlates strongly with the observed performance gap (r = 0.81), validating the theoretical predictor.

Scalability: Performance improvements hold for K in {2, 4, 8} basis policies; gains plateau at K = 8.

ABLATIONS AND INTERPRETATION

Routing vs. random assignment: Random assignment reduces hypervolume by 7%, confirming conflict-aware routing is the key driver.

Attribution vs. uniform sampling: Replacing kernel attribution with uniform sampling costs ~20% in sample efficiency.

Conflict coefficient threshold: Task suites with c_max < 0.1 show no significant gain from ensembles—a practical criterion for when ensembles are needed.

REFERENCE

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